

Task 00: Example Workflow

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1 Goal

This task demonstrates the full workflow of the repository: code $\rightarrow$ plots $\rightarrow$ report. The goal is to show how a weekly task can be organized, how figures are generated with Python, how they are included in $\text{\LaTeX}$, and where the final PDF is saved.

2 Generating Plots

Plots are created using Python and saved as PDF files in

`../build/`

The plotting style is controlled by the shared matplotlib configuration file

`latex/matplotlibrc`

This keeps fonts, figure sizes, and general plot formatting consistent across all tasks.

2.1 Example Function

Figure 1 shows a simple example plot generated from `code/make_plots.py`.

2.2 Training Curve

Figure 2 shows an example of a plot that could represent a training loss curve.

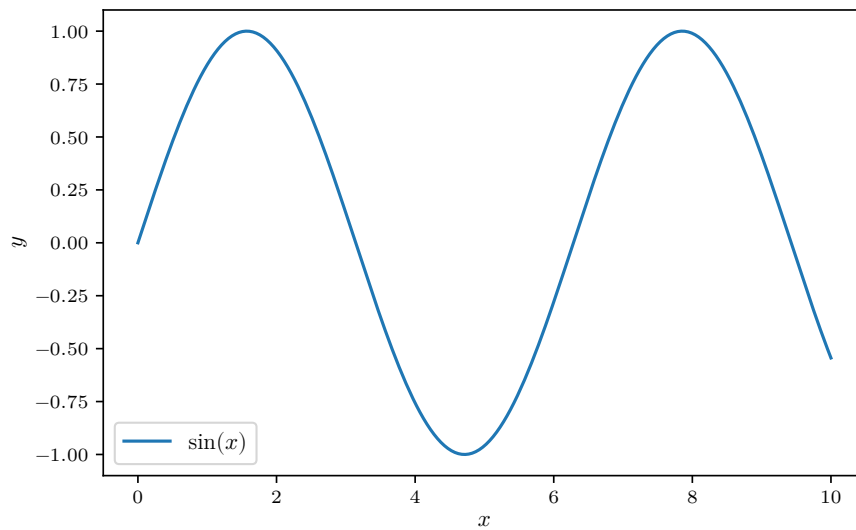


Figure 1: Sine plot generated with Python.

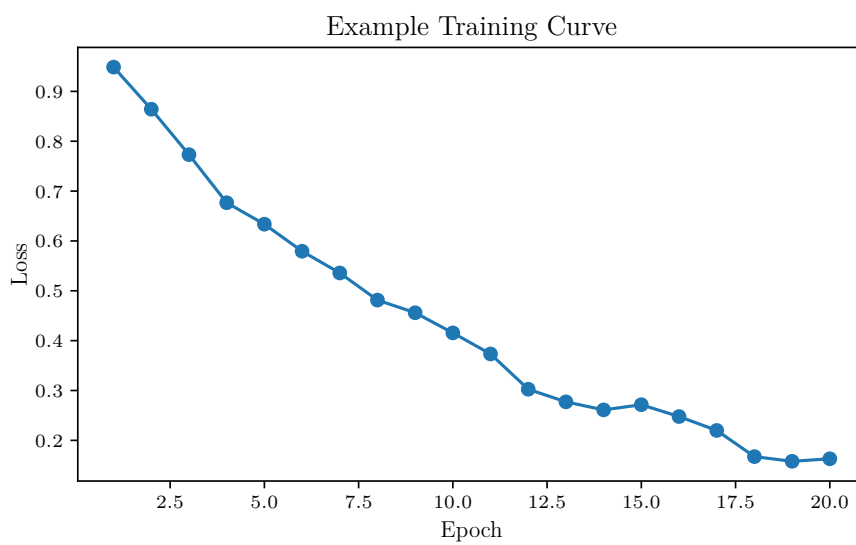


Figure 2: Example training loss curve.

3 Bibliography

References are stored globally in

`latex/references.bib`

They are loaded through the shared header file. This means every task can use the same bibliography database.

For example, a source can be cited with

`\cite{example-key}`

or, depending on the context,

`\autocite{example-key}`

The bibliography is printed at the end of the report with

`\printbibliography`

4 Build Output

The task is built using `make`. Running

`make`

inside the task folder first generates the figures and then compiles the report.

Temporary build files are written to

`build/`

The final PDF is copied to

`report/main.pdf`

This is the file that should be opened or submitted.

5 How it Works

- Python scripts are stored in `code/`
- Plots are generated by `code/make_plots.py`
- Figures are saved in `../build/`
- The report is written in `report/main.tex`
- External PDF figures are included with `\includegraphics`
- Bibliography entries are stored in `latex/references.bib`
- Temporary files are collected in `build/`
- The final report is saved as `report/main.pdf`

6 Conclusion

This setup allows reproducible experiments and clean reports. Each task stays self-contained, while shared settings such as the $\text{\LaTeX}$ header, bibliography, and matplotlib configuration are reused across the repository.